Lesson 2-8 · Session 1

# Finding the Mean

Name _____ Date _____ Period _____

Maryland Standard 6.SP.B.5c · Learning target — I can find the mean of a data set by sharing the total equally among the values.
Show a model, calculation, or explanation. Answer every part and label units when needed.

Exercises 1–12 earn 1 point for a correct response and 1 for supporting work. Spiral Review and Test Practice earn 1 point each.

## Learn from a model

**REMEMBER**

- The mean (average) equals the sum of all values divided by the number of values.
- The mean can be affected a lot by very high or very low values.
- Watch out — every value counts, including repeats and zeros.
- Check it — the mean must land between the least and the greatest value.

**WORKED EXAMPLE**

Finding the Mean of a Data Set Five afternoons of dog walking earned 14, 21, 17, 19, and 24 dollars. Find the mean.

**Step 1** Add every value. The five amounts give a total of 95 dollars.

**Step 2** Count the values. There are five of them.

**Step 3** Share the total evenly: divide it by the count of values.

**Step 4** Check: the mean should land between the least and the greatest value.

**Answer The mean is 19 dollars — what each afternoon would earn if the money were shared evenly.**

## Practice

**1.** Find the mean of: 4, 8, 6, 10, 7

Sum _____ divided by _____ values Mean _____

Answer _____

SHOW YOUR THINKING

**2.** Five test scores are 82, 90, 76, 88, 94. Find the mean score.

Answer _____

SHOW YOUR THINKING

**3.** If one more score of 100 is added to problem 2's data (now 6 scores), will the mean go up or down? Explain without calculating.

Answer _____

EXPLAIN WITH MATHEMATICAL EVIDENCE

**4.** Find the mean of each set.

a. 3, 5, 7 _____
b. 10, 14, 12, 8 _____
c. 20, 25, 30 _____
d. 6, 6, 9, 11 _____
e. 40, 50, 45, 65 _____

Answer _____

SHOW YOUR THINKING

**5.** Ticket Money. Five families paid $18, $24, $15, $21, and $22 for tickets. Find the mean amount.

Answer _____

SHOW YOUR THINKING

**6.** Plant Watering. Six plants were given 12, 9, 15, 11, 14, 17 milliliters of water. Find the mean amount and explain what it means.

Answer _____

EXPLAIN WITH MATHEMATICAL EVIDENCE

**7. One Big Value. Four students collected 8, 10, 12, 50 cans. Find the mean and explain why it sits above three of the four values.**
**Sentence Frame: "I know this because..."**

Answer _____

EXPLAIN WITH MATHEMATICAL EVIDENCE

**8.** Must the Mean Appear? Amari says the mean of a data set must always be one of the values in the set. Give a data set that shows he is wrong and explain.

Answer _____

EXPLAIN WITH MATHEMATICAL EVIDENCE

**9.** Find the mean of 11, 15, 19, and 23.

Answer _____ mean

SHOW YOUR THINKING

**10.** Find the mean of 7, 12, 9, 14, 18, and 12.

Answer _____ mean

SHOW YOUR THINKING

**11.** Test Average. A student's five test scores total 410 points. Find the mean score.

Answer _____ points

SHOW YOUR THINKING

**12.** Find the Mistake. Kai found the mean of 5, 9, and 13 by dividing 27 by 13. Explain the mistake and give the correct mean.

Answer _____ the mistake and the mean

**EXPLAIN WITH MATHEMATICAL EVIDENCE**

## Spiral review

**13.** From Lesson 2-7 Session 2. Find 9.9 ÷ 0.9.

Answer _____

**SHOW YOUR THINKING**

**14.** From Lesson 2-5 Session 1. Find the range of 14, 39, 22, 31.

Answer _____

**SHOW YOUR THINKING**

**15.** From Lesson 2-3 Session 2. Find the median of 10, 16, 22, 28.

Answer _____

**SHOW YOUR THINKING**

## Test practice

**16.** Find the mean of 3, 7, 11.

| | |
|---|---|
| □ A. 7 | □ B. 6 |
| □ C. 21 | □ D. 11 |

**17.** Four values total 96. The mean is —

| | |
|---|---|
| □ A. 26 | □ B. 24 |
| □ C. 384 | □ D. 92 |

**18.** The mean is sometimes called the —

| | |
|---|---|
| □ A. range | □ B. median |
| □ C. average | □ D. mode |

## Check and revise

| CHECK YOUR MATHEMATICS | HOW DID IT GO? |
|---|---|
| Include every data value, including zero, in the count.<br>Every part answered / Work shown / Units labeled when needed | I can do this:<br>Independently / With a hint / With help<br>Question to revisit: _____ |

One correction or question I want to discuss: _____

Check one response against the data. State which measure or feature supports your answer.